

EFFICIENT APPROXIMATION OF EXTREMAL FIELD MAPS USING MESH-FREE INTERPOLATION METHODS

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For spacecraft guidance and control, the ability to quickly re-optimize a trajectory is crucial. Indirect methods for solving optimal control problems (OCPs) are widely used for trajectory optimization. In indirect methods, the necessary information to describe the optimal solution is contained entirely within the initial state, optimal initial costates, and the differential equations of motion. For any OCP, a range of optimal solutions can be pre-computed at discrete sets of parameters and efficiently stored as an Extremal Field Map (EFM). Using interpolation, the EFM provides optimal solutions for a continuous, finite domain parameter space. In this work, extremal field maps are constructed efficiently using a reverse integration approach and interpolated using a polyharmonic spline interpolation method. Mappings generated using this method achieve an error magnitude at interpolation points which nearly satisfy the necessary conditions for optimality, without the need for re-computation.

INTRODUCTION

In the optimization of spacecraft maneuvers, there is an ever-present level of uncertainty in both the position and velocity states of the spacecraft and the parameters which describe it. Over the course of a mission, this uncertainty compounds forcing engineers to redesign the trajectory to account for updated information. Due to the long duration of many missions, these trajectory re-designs may be required numerous times over the mission lifetime, costing engineers valuable time and resources. To prepare for this inevitability, a range of optimized trajectories which neighbor the nominal trajectory can be pre-computed and stored and used to quickly re-optimize the trajectory. Using an indirect optimal control approach, the state and costate differential equations can recover the entire optimal trajectory and control history given only the state and optimal costates. This fact can be used to construct an Extremal Field Map (EFM) which relates a set of possible states neighboring the nominal trajectory to the initial costates which optimize the remainder of the trajectory. Using an appropriate interpolation scheme, the map can be used to quickly recover the optimal trajectory and control history for any set of parameters within a continuous, defined range of parameters. This approach yields a quasi-closed loop guidance system which can be used to quickly re-optimize the trajectory given a new set of initial conditions. With the proper construction, EFMs can be implemented as autonomous guidance systems for spacecraft.¹

Extremal field maps can be constructed using a variety of methods. Trivially, indirect optimal control solutions can be computed at a set of nodes, from which an interpolation can be created. Us-

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ing orthogonal polynomial series to generate the optimal trajectories using collocation improves the computation time, and secondarily to create a smooth interpolation between extremal field points to approximate the optimal costates at interior points without needing to solve a nonlinear program.² Alternatively, the implicit function theorem can be leveraged to efficiently compute neighboring trajectories on a uniform grid, and powerful meshed approximation methods such as global-local orthogonal polynomial weight function (GLO-MAP) approximation can be used to interpolate a small parameter space.³ However, these methods require repeatedly solving a nonlinear program for the optimal costate vector, requiring significant up-front computation time. Alternatively, for rendezvous problems - where the full final state vector is known - optimal trajectories can be generated through propagation alone by integrating backwards in time from the final state with an arbitrary costate vector.⁴ This method drastically reduces the necessary computation time for constructing an EFM, requiring only a single propagation per data point, rather than a full nonlinear program. The downside to this reverse integration method is that, for nonlinear systems, the extremal node points do not exist on any sort of regular grid, so a mesh-free approximation method is necessary to interpolate the data. In this work, extremal field maps will be efficiently constructed using the reverse integration method described, and interpolated using a polyharmonic interpolation kernel using radial-basis functions with a polynomial component.

METHODOLOGY

For most trajectory optimization problems, indirect methods can be used. The most general form of an optimal control problem (OCP) involves minimizing a cost functional J with respect to a number of dynamic constraints $\mathbf{f}$ and boundary conditions ψ .⁵

$$J = \phi + \int_{t_0}^{t_f} L dt \quad (1)$$

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}, t) \quad (2)$$

$$\psi(\mathbf{x}(t_f), t_f) = \mathbf{0} \quad (3)$$

From this, the Hamiltonian can be constructed, and from it the necessary conditions contain the costate differential equations and the optimality conditions. Additionally, the optimal control $\mathbf{u}^*$ can be found, typically as a function of the costates, using the necessary conditions or Pontryagin's principle.⁶

$$H = L + \boldsymbol{\lambda}^T \mathbf{f} \quad (4)$$

$$\dot{\boldsymbol{\lambda}} = -\frac{\partial H}{\partial \mathbf{x}} \quad (5)$$

$$\mathbf{u}^* \in \arg \min H \quad (6)$$

Using this information, the problem becomes a boundary value problem (BVP), which can be solved using a variety of methods. In this work, a single shooting method using numerical integration is used. For a particular problem, a nominal trajectory with a defined initial state $\mathbf{x}_{0_{nom}}$ and final state $\mathbf{x}_{f_{nom}}$ can be defined. Then, a set of N parameters $\mathbf{p}$ containing a selection of initial conditions, final conditions, and/or characteristic constants from the differential equations, can be constructed to be varied over some finite parameter space around the nominal trajectory. Then, an extremal field map defined as,

$$\Lambda : \mathbf{p}_{N \times 1} \rightarrow \boldsymbol{\lambda}_{0_{M \times 1}}^*, \quad (7)$$

is created by computing the M optimal initial costates λ_0^* for OCPs at a set of discrete points in the parameter space.

To create a continuous mapping, an interpolation method which can handle mesh-free, unstructured data is needed. Radial basis function (RBF) interpolation is a powerful tool for this purpose. RBFs are a class of real-valued functions for which the output is a function of only the distance $r = \|\mathbf{x} - \mathbf{c}_i\|$ between the interpolation point and the center $\mathbf{c}_i$. Relevant examples of radial basis functions include the gaussian and inverse quadratic functions which are strictly positive definite but require a hyperparameter tuning, as well as the families of polyharmonic splines and compactly supported radial basis functions.⁷ Since the data generated for use in this work is mostly irregular, the thin plate spline kernel (a special case of polyharmonic splines),

$$\phi(r) = r^2 \ln(r) = r \ln(r^r), \quad (8)$$

is chosen for its absence of a tuning parameter numerical stability in interpolation applications.⁸ With a kernel chosen, a multivariate least squares fit can be created by solving the linear system

$$\mathbf{f} = \Phi \boldsymbol{\alpha} = \begin{bmatrix} \phi(\|\mathbf{x}_1 - \mathbf{x}_1\|) & \phi(\|\mathbf{x}_1 - \mathbf{x}_2\|) & \dots & \phi(\|\mathbf{x}_1 - \mathbf{x}_N\|) \\ \phi(\|\mathbf{x}_2 - \mathbf{x}_1\|) & \phi(\|\mathbf{x}_2 - \mathbf{x}_2\|) & \dots & \phi(\|\mathbf{x}_2 - \mathbf{x}_N\|) \\ \vdots & \vdots & \ddots & \vdots \\ \phi(\|\mathbf{x}_N - \mathbf{x}_1\|) & \phi(\|\mathbf{x}_N - \mathbf{x}_2\|) & \dots & \phi(\|\mathbf{x}_N - \mathbf{x}_N\|) \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_N \end{bmatrix}, \quad (9)$$

$$\mathbf{f}(\mathbf{x}) \approx \sum_{i=1}^N \alpha_i \phi(\|\mathbf{x} - \mathbf{x}_i\|) \quad (10)$$

for the weight vector $\boldsymbol{\alpha}$, where $\mathbf{f}$ is a vector of data points, and $\mathbf{f}(\mathbf{x})$ is the interpolation function. While radial basis function interpolation has many advantages, it suffers from the fact that the interpolation has no information about the true global shape of the data, only the values at specified distances from the centers of each radial basis function. This means the accuracy of the fit is very dependent on the structure of the data, and can often be very sensitive to the spacing between centers. To combat this issue, polyharmonic splines can be combined with a polynomial term to fit the data at both global and local scales.⁹ This can be implemented by constructing a Vandermonde matrix for a scalar function of a vector polynomial of order 2 such that,

$$\mathbf{B} = \begin{bmatrix} 1 & \mathbf{x}_1^T & \mathbf{x}_1^T \otimes \mathbf{x}_1^T \\ 1 & \mathbf{x}_2^T & \mathbf{x}_2^T \otimes \mathbf{x}_2^T \\ \vdots & \vdots & \vdots \\ 1 & \mathbf{x}_N^T & \mathbf{x}_N^T \otimes \mathbf{x}_N^T \end{bmatrix} = \begin{bmatrix} 1 & \mathbf{x}_1^T & x_{1_1} \mathbf{x}_1^T & x_{1_2} \mathbf{x}_1^T & \dots x_{1_d} \mathbf{x}_1^T \\ 1 & \mathbf{x}_2^T & x_{2_1} \mathbf{x}_2^T & x_{2_2} \mathbf{x}_2^T & \dots x_{2_d} \mathbf{x}_2^T \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & \mathbf{x}_N^T & x_{N_1} \mathbf{x}_2^T & x_{N_2} \mathbf{x}_2^T & \dots x_{N_d} \mathbf{x}_2^T \end{bmatrix} \quad (11)$$

where $(\otimes)$ denotes the Kronecker product, which is simply shorthand for the expansion of the general 2nd degree scalar polynomial for a d dimensional vector.¹⁰ Combining this matrix with the radial basis function system, a least squares problem can be formulated such that,

$$\begin{bmatrix} \Phi & \mathbf{B} \\ \mathbf{B}^T & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{w} \\ \mathbf{v} \end{bmatrix} = \begin{bmatrix} \mathbf{f} \\ \mathbf{0} \end{bmatrix} \quad (12)$$

which has a unique solution so long as $\mathbf{B}$ is full rank. In practice, the rank of $\mathbf{B}$ can be checked during construction, and if it happens to be rank deficient, its columns can be reduced until it is full

rank. This is only necessary when the ratio of node points to columns of the Vandermonde matrix is low. The additional constraint $B^T \mathbf{v} = 0$ enforces orthogonality between the radial basis function weights and the polynomial basis functions. Conveniently, these constraints can be eliminated by leveraging the null space of the Vandermonde matrix, which can be computed using a QR decomposition. The linear system can then be solved in two steps. First, compute Z , which is an orthonormal basis for the null space of B such that $B^T Z = 0$ and $\mathbf{w} = Z\gamma$ for some vector γ . Substituting this into the first row of the system shown previously yields,

$$\Phi \mathbf{w} + B \mathbf{v} = \Phi Z \gamma + B \mathbf{v} = \mathbf{f} \quad (13)$$

$$Z^T \Phi Z \gamma = Z^T \mathbf{f} \quad (14)$$

from which the vector γ can be computed. Second, the original weights $\mathbf{w}$ and polynomial coefficients $\mathbf{v}$ can be recovered by solving

$$\mathbf{w} = Z \gamma \quad (15)$$

$$B \mathbf{v} = \mathbf{f} - \Phi \mathbf{w} = \mathbf{f} - \Phi Z \gamma. \quad (16)$$

This method improves the conditioning of the linear system by removing the polynomial basis functions from the radial basis function interpolation. The resulting system, $Z^T \Phi Z$, is also symmetric positive definite, allowing for an efficient solution method leveraging Cholesky decomposition.⁹

Using this polyharmonic spline interpolation method with data generated via the reverse integration method, extremal field maps can be constructed efficiently. This data structure takes the shape of a bundle of trajectories surrounding the nominal, all leading to the final boundary condition. In the construction of the data set, it is important to use sets of perturbed final costate which are evenly distributed to improve the conditioning of the linear system. For systems where the costate-space is two dimensional, a fibonacci spiral can be used to cover the space with an efficient packing.¹¹ In higher dimensions, algorithms such as Lloyd's algorithm or numerical variations of Thompson's problem which generate evenly spaced sets of points can be used to perturb the nominal final costate and generate kernel data which is roughly evenly spaced in phase space.^{12,13} Additionally, it may be beneficial to normalize the parameter node values before creating the interpolation to improve the conditioning.

APPLICATIONS

To validate the extremal field map generation methods described, consider the minimum energy optimal control problem for a planar two-body orbit transfer. The classic Earth to Mars rendezvous problem is used, ignoring third body perturbations. The boundary conditions,

$$\mathbf{r}_0 = [-140699693, -51614428]^T \text{ (km)} \quad (17)$$

$$\mathbf{v}_0 = [9.774596, -28.07828]^T \text{ (km/s)} \quad (18)$$

$$\mathbf{r}_f = [-172682023, 176959469]^T \text{ (km)} \quad (19)$$

$$\mathbf{v}_f = [-16.427384, -14.860506]^T \text{ (km/s)} \quad (20)$$

are referenced from Taheri et al. 2018.¹⁴ The dynamics of the system are sun-centric, such that,

$$\begin{bmatrix} \dot{\mathbf{r}} \\ \dot{\mathbf{v}} \end{bmatrix} = \begin{bmatrix} \mathbf{v} \\ -\frac{\mu}{\|\mathbf{r}\|^3} \mathbf{r} + \mathbf{u} \end{bmatrix} \quad (21)$$

for which the cost functional, Hamiltonian, and optimal control are given by,

$$J = \frac{1}{2} \int_{t_0}^{t_f} \mathbf{u}^T \mathbf{u} dt, \quad H = \frac{1}{2} \mathbf{u}^T \mathbf{u} + \lambda_r^T \mathbf{v} + \lambda_v^T \left(-\frac{\mu}{\|\mathbf{r}\|^3} \mathbf{r} + \mathbf{u} \right) \quad \mathbf{u}^* = -\lambda_v \quad (22)$$

which have the corresponding costate dynamics,

$$\begin{bmatrix} \dot{\lambda}_r \\ \dot{\lambda}_v \end{bmatrix} = \begin{bmatrix} \frac{\mu}{\|\mathbf{r}\|^3} \lambda_v - \frac{3\mu}{\|\mathbf{r}\|^5} (\lambda_v^T \mathbf{r}) \mathbf{r} \\ -\lambda_r \end{bmatrix} \quad (23)$$

which are derived from the necessary conditions.⁵ The nominal trajectory is generated by solving the two point boundary value problem using a shooting method with the boundary constraint vector,

$$\boldsymbol{\psi} = \begin{bmatrix} \mathbf{r}(t_f) - \mathbf{r}_f \\ \mathbf{v}(t_f) - \mathbf{v}_f \end{bmatrix}, \quad (24)$$

such that for the nominal trajectory, the nominal final costate values,

$$\boldsymbol{\lambda}_{f_{nom}} = [0.0327280 \quad -0.0900747 \quad -0.000539642 \quad 0.144939]^T, \quad (25)$$

are perturbed by a set of 4000 pseudo-evenly spaced vectors generated using Lloyd's algorithm.¹² Lloyd's algorithm is also used to generate a range of times for which the perturbed final costate vectors can be propagated, generating a set of off nominal trajectories which are used to construct the extremal field map. The mapping,

$$\Lambda : [\mathbf{r}^\top \quad \mathbf{v}^\top \quad t] \rightarrow [\boldsymbol{\lambda}_r^\top \quad \boldsymbol{\lambda}_v^\top], \quad (26)$$

is 5-to-4, taking the state and time-to-go as inputs and returning the corresponding optimal costates as outputs. The interpolation kernel uses a thin-plate-spline basis with a 2nd-order Vandermonde matrix, in an attempt to match the order of the costate dynamics. The resulting data set, as shown in figure 1

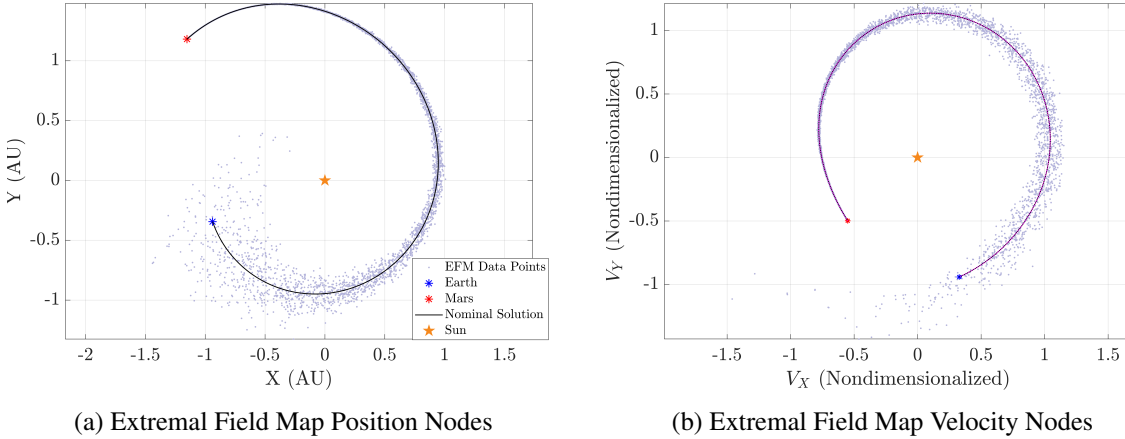


Figure 1: Extremal Field Map Data Set

forms a bundle of trajectories around the nominal trajectory, providing optimal guidance for any spacecraft within the domain. To validate the extremal field map, 100 validation trajectories are generated by sampling initial conditions from between EFM node points, then using the map to recover

the initial costates. These trajectories are then propagated forward in time to the final boundary condition, and the rendezvous constraint is evaluated. Figure 2a shows the generated validation trajectories, which have an overall constraint error norm of $\|\psi\| = 1.6891 \times 10^{-5}$ with a mean absolute error of $\psi = 3.7215 \times 10^{-7}$ when using non-dimensionalized units. If viewed in dimensional units, the mean position error of the boundary condition is approximately 50 km, and the mean velocity error is on the order of 10^{-6} km/s, which for an interplanetary rendezvous problem is very small. This magnitude of the constraint errors is directly correlated to the time-to-go of the validation trajectories as illustrated in figure 2b, as longer propagation times are more sensitive to the initial conditions; however, this indicates that the extremal field map, if implemented as a quasi-closed loop guidance system, could improve the convergence with additional EFM sampling over the course of the trajectory. Additionally, even the largest error magnitudes would be considered solutions or very near solutions in a nonlinear solver such as Matlab's `fsolve`, which sets its tolerance based on the error norm squared.

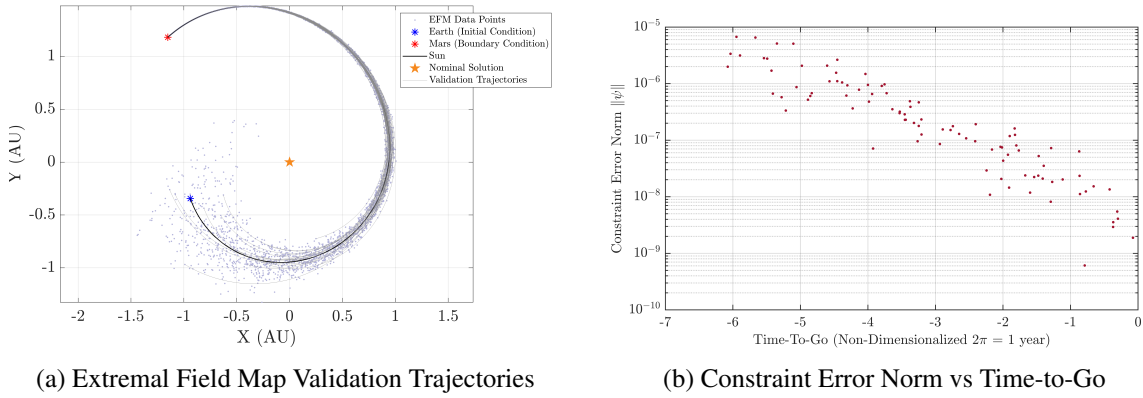


Figure 2: Extremal Field Map Validation Trajectories and Error Norms

CONCLUSION

The results presented in this work demonstrate the feasibility of constructing extremal field maps without the need to solve a large number of nonlinear programs, replacing that process with a single propagation per data point. The trade-off for this efficiency is the need for a mesh-free approximation method which can handle the unpredictability of costate data. The polyharmonic spline interpolation method not only provides a smooth and accurate approximation of the data, but its multiresolution nature provides both a global shape and local detail to the approximation, allowing for a more accurate mapping. By removing the computational burden of solving thousands of nonlinear programs and trading the complexity of training a more complicated interpolator, this method provides a more efficient and effective way to construct extremal field maps for practical applications. Future investigation into applications of this method to more complex systems, such as the minimum-time and minimum-fuel problems in the full 3 dimensions are necessary to validate this method for realistic use cases. However, the potential for this method to be used as a quasi-closed loop guidance system provides a significant impetus for further research into the generation of extremal field maps and their applications in trajectory optimization.

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